

ARKHARA, Russian Federation

WMO: 315940

Lat: 49.417N

Lon: 130.083E

Elev: 442

StdP: 14.46

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.5	-30.7	-38.4	0.6	-34.1	-34.9	0.8	-30.2	14.5	-0.5	12.2	-1.4	1.0	320	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	85.9	72.4	83.0	70.5	80.5	68.9	74.5	82.9	72.6	80.5	70.9	78.2	5.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	119.6	79.7	69.9	111.8	77.7	68.2	105.2	75.6	38.5	83.1	36.8	80.6	35.2	78.5	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.2	14.1	11.8	-38.8	91.3	5.1	2.6	-42.5	93.1	-45.5	94.7	-48.4	96.1	-52.1	98.0		
(5)			-38.9	78.7	5.1	2.6	-42.6	80.6	-45.5	82.1	-48.4	83.6	-52.1	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	-12.7	-3.2	16.1	39.2	54.5	65.1	70.8	66.7	54.5	37.3	11.9	-9.4		
	DBStd	30.96	8.97	10.34	12.71	8.90	7.57	6.15	4.61	5.69	7.96	9.48	13.28	9.80		
	HDD50	8300	1943	1489	1050	343	39	0	0	1	44	407	1142	1842		
	HDD65	12154	2408	1909	1515	775	338	73	7	46	325	859	1592	2307		
	CDD50	2003	0	0	0	18	177	453	645	518	178	14	0	0		
	CDD65	382	0	0	0	1	11	75	187	99	9	0	0	0		
	CDH74	2829	0	0	0	16	158	643	1293	659	59	1	0	0		
	CDH80	657	0	0	0	2	28	168	333	121	5	0	0	0		
	WSAvg	4.6	3.1	4.0	5.2	6.4	6.1	4.4	4.0	3.8	4.4	5.5	4.8	3.4		
	PrecAvg	25.2	0.3	0.3	0.4	1.3	2.1	3.9	5.4	5.6	3.4	1.3	0.7	0.5		
	PrecMax	33.1	0.7	0.9	1.5	3.2	3.9	7.4	11.0	9.3	8.6	3.9	2.1	1.2		
(15) Precipitation	PrecMin	17.3	0.0	0.0	0.0	0.0	0.5	0.6	1.4	2.5	1.3	0.0	0.2	0.1		
	PrecStd	3.7	0.2	0.2	0.3	0.7	0.9	1.9	2.3	1.9	1.8	0.9	0.5	0.3		
	0.4%	DB	17.7	31.1	50.2	76.2	84.0	89.7	90.3	86.5	79.6	69.1	46.7	24.6		
	MCWB	15.5	26.1	38.9	55.4	62.0	71.1	74.4	74.6	67.5	53.6	38.4	21.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	11.5	24.9	44.1	65.2	78.8	84.5	86.5	83.4	75.2	61.6	40.0	16.0		
	MCWB	9.9	21.3	35.3	48.8	59.4	69.0	73.5	72.7	63.7	49.6	34.7	14.2			
	5%	DB	6.6	19.9	39.1	59.5	73.9	80.9	83.8	80.7	71.8	56.8	35.2	11.1		
	MCWB	5.2	17.3	32.3	45.7	57.1	67.3	72.4	71.0	61.6	46.4	31.1	9.7			
	10%	DB	2.6	14.4	35.2	54.5	69.4	77.4	81.0	77.7	68.0	52.3	31.3	6.7		
	MCWB	1.5	12.2	29.7	42.6	55.1	65.8	70.7	69.0	59.4	43.7	28.4	5.5			
	0.4%	WB	15.5	26.5	39.7	56.9	64.5	74.9	77.3	76.7	69.1	55.4	39.5	21.6		
	MCDB	17.4	30.7	48.5	74.8	79.3	85.9	86.4	83.9	77.1	65.7	44.9	24.0			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	9.9	21.5	35.7	50.2	61.2	71.1	74.9	74.1	65.9	51.6	34.9	14.4		
	MCDB	11.4	24.8	42.8	62.2	74.1	81.6	83.8	81.1	73.2	60.1	39.2	15.8			
	5%	WB	5.3	17.3	32.7	46.4	58.7	68.8	73.3	72.0	63.3	47.7	31.7	9.8		
	MCDB	6.5	20.0	38.6	57.4	71.1	78.4	81.8	78.9	69.6	55.0	34.8	11.1			
	10%	WB	1.6	12.5	30.0	43.2	56.4	66.8	71.7	70.1	60.7	44.4	28.1	5.6		
	MCDB	2.5	14.4	35.0	53.5	67.5	75.6	79.1	76.3	66.1	51.6	31.5	6.7			
(35) Mean Daily Temperature Range	MDBR	21.6	24.5	22.6	20.6	21.6	19.1	16.9	17.9	21.6	20.1	18.6	19.8			
	5% DB	MCDBR	22.4	25.1	20.9	28.2	29.0	25.2	21.7	21.4	24.5	26.6	19.2	20.2		
	MCWBR	21.0	22.1	14.0	15.8	14.6	12.7	11.3	11.8	14.0	15.6	14.3	18.9			
	5% WB	MCDBR	22.3	24.6	19.8	25.5	24.8	21.9	19.2	18.8	20.8	22.8	17.2	20.0		
	MCWBR	20.9	21.7	13.4	15.1	13.6	12.1	10.8	11.1	12.9	14.5	13.2	18.8			
(40) Clear-Sky Solar Irradiance	taub	0.283	0.314	0.391	0.537	0.581	0.512	0.490	0.442	0.410	0.424	0.350	0.289			
	taud	2.128	2.037	1.889	1.723	1.714	1.977	2.094	2.219	2.215	2.000	2.069	2.099			
	Ebn at Noon	236	255	251	225	222	241	245	252	246	209	201	211			
	Edh at Noon	26	36	51	68	71	55	48	41	37	38	28	23			
(44) All-Sky Solar Radiation	RadAvg	564	954	1420	1548	1662	1787	1667	1494	1206	793	531	438			
	RadStd	15	21	87	161	146	226	138	108	117	56	38	22			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page